

The University of Lahore
Department of Computer Science & IT
Final term Examination



Fall-2018

Course Title:	Programming Fundamentals-I	Course Code:	CS02101	Credit Hours:	4
Course Instructors:	Dr. Muhammad Adnan Hashmi Dr. Abbas Khalid Mr. Imran Riaz Mr. Rehan Ahmed Mr. Adeem Ali Anwar Mr. Waqas Arshad Mr. Saddam Hussain	Program Name:	BSCS		
Time Allowed:	120 Minutes	Maximum Marks:	50		
Date:	17-01-2019	Course Mentor:	Dr. Abbas Khalid		
Student's Name:		Signature:			
Reg. No:		Section:			
Important Instructions / Guidelines: - Understanding of the questions is the part of the question paper so do not disturb the invigilators. - Make sure that you have 4 typed pages including this page. - Kindly leave your mobile phones outside exam hall. - Solve questions 1, 2, 3 on question paper and question 4 on answer sheet provided separately. - Your paper will be marked as ZERO, if involved in any kind of cheating.					

Question#1: Encircle the correct option.

[10 X 1 = 10 Marks]

1. What is output? `int a[]={1,2,3,4,5}; cout << a;`
 - a) 12345
 - b) address of a
 - c) garbage value
 - d) None of these
2. Which of the following is not a valid C++ expression for adding 5 in A?
 - a) `A = 5+A`
 - b) `A += 5`
 - c) `A =+ 5`
 - d) `A = A+5`
3. Which of the following is valid variable name?
 - a) long
 - b) int
 - c) float
 - d) Double
4. Number of bytes occupied by the following array: `char arr [] = {'A', 'B', 'C', 'D'};`
 - a) 4
 - b) 5
 - c) 8
 - d) 10
5. Conditional operator is an alternate of:
 - a) if
 - b) if-else
 - c) nested if
 - d) None
6. Which of the following is not a valid case statement in a switch?
 - a) case 1:
 - b) case 'A':
 - c) case 1.5:
 - d) None

7. Which of the following is equivalent to $(p \geq q)$?
- a) $!p < q$ b) $p < q$ c) $p > q$ d) $!(p < q)$
8. How many times the given code prints "OOP"? `int a = 0; while (a++ < 10) {cout << "OOP";}`
- a) 9 b) 10 c) 11 d) 12
9. If you want a user to enter exactly 10 values, which loop would be the best to use?
- a) for b) while c) do-while d) None
10. What is output? `int arr [4] = {1,2}; cout << arr [2];`
- a) 2 b) -1 c) 0 d) 1

Question # 2: Write down output of the following code segments. [3 X 4 = 12 Marks]

CODE	OUTPUT
a) <pre> void main () { char grade = 'B'; switch(grade) { case 'A': cout << "Grade point is 4" << endl; break; case 'B': cout << "Grade point is 3" << endl; case 'C': cout << "Grade point is 2" << endl; case 'D': cout << "Grade point is 1" << endl; case 'F': cout << "Grade point is 0" << endl; break; default: cout << "Invalid grade"; } } </pre>	
b) <pre> void main () { int m = 16, n =2; while (n <= 100) { n = 2*n + m; if (m%2 == 0) m = m/2; cout << n << " " << m << endl; } } </pre>	

c) <pre> void main () { int arr [10]; for (int i = 1; i <= 10; i++) { arr [i - 1] = 5 + i; } for (int i = 4; i < 8; i++) { cout << arr[i] << " "; } } </pre>	
--	--

Question # 3: Find out syntax errors in the given code and rewrite the correct code.

[2 X 4 = 8 Marks]

Incorrect Code	Correct Code
a) <pre> void main () { int w, x = "5", y, z = "3"; y = 2 (z + 3); w = x - 2 * y + z; if (x => y) x++; if else (x<10 & y>20) w++; cout<<w<<" "<<x<<" "<<y<<" "<<z; } </pre>	
b) <pre> void main { int s = 10; int n = 10; Do { s=s++n; if (s<10 n>10) breaks; } cout<<a; while (s! = 20); } </pre>	

Question # 4: Programming Questions:**[4+8+8 = 20 Marks]**

a). Write a C++ program that inputs the value of x and solves the following equation.

$$3x^2+6x+4$$

c). Write a program to display the following output using any nested loop.

0

5 10

15 20 25

30 35 40 45

50 55 60 65 70

75 80 85 90 95 100

d). Write a program that inputs marks of 50 students in an integer array. The program displays the number of students in each grade. The criteria are as follows:

Marks	Grade
90 or above	A
70 to 89	B
50 to 69	C
Below 50	F